Dictonary

Dictionaries are used to store data values in key:value pairs.

A dictionary is a collection which is ordered*, changeable and do not allow duplicates.

1.create dictonary

In [4]:
```
data = {"name":"Reeta","age":20}
print(data)
```

```
{'name': 'Reeta', 'age': 20}
```

2.find len

In [7]:
```
data = {"name":"Reeta","age":20}
print(len(data))
```

```
2
```

3.data type

In [10]:
```
data = {"name":"Reeta","age":20}
print(type(data))
```

```
<class 'dict'>
```

4.dict constructor

In [13]:
```
data=dict(name="Reeta",age=20)
print(data)
```

```
{'name': 'Reeta', 'age': 20}
```

5.access item

In [20]:
```
data = {"name":"Reeta","age":20}
print(data["name"])
```

```
Reeta
```

In [22]:
```
data = {"name":"Reeta","age":20}
a=data["name"]
```

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```
print(a)
```

```
Reeta
```

using get

In [25]:

```
data = {"name":"Reeta","age":20}
a=data.get("name")
print(a)
```

```
Reeta
```

6.access all keys

In [28]:

```
data = {"name":"Reeta","age":20}
a=data.keys()
print(a)
```

```
dict_keys(['name', 'age'])
```

7.access all values

In [31]:

```
data = {"name":"Reeta","age":20}
a=data.values()
print(a)
```

```
dict_values(['Reeta', 20])
```

8.get all data

In [34]:

```
data = {"name":"Reeta","age":20}
a=data.items()
print(a)
```

```
dict_items([('name', 'Reeta'), ('age', 20)])
```

9.Check if Key Exists

In [37]:

```
data = {"name":"Reeta","age":20}
if "name" in data:
    print("yes")
```

```
yes
```

In [45]:

```
data = {"name":"Reeta","age":20}
if "year" not in data:
    print("yes")
else:
    print("no")
```

```
yes
```

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10.change item

In [48]:

```
data = {"name":"Reeta","age":20}
data["age"]=21
print(data)
```

```
{'name': 'Reeta', 'age': 21}
```

In [54]:

```
data = {"name":"Reeta","age":20}
data.update({"age":21})
print(data)
```

```
{'name': 'Reeta', 'age': 21}
```

11.add item

In [57]:

```
data = {"name":"Reeta","age":20}
data["year"]=2005
print(data)
```

```
{'name': 'Reeta', 'age': 20, 'year': 2005}
```

In [59]:

```
data = {"name":"Reeta","age":20}
data.update({"year":2005})
print(data)
```

```
{'name': 'Reeta', 'age': 20, 'year': 2005}
```

12.remove item

pop()

In [65]:

```
data = {"name":"Reeta","age":20}
data.pop("age")
print(data)
```

```
{'name': 'Reeta'}
```

popitem() - it remove last value

In [68]:

```
data = {"name":"Reeta","age":20}
data.popitem()
print(data)
```

```
{'name': 'Reeta'}
```

del

In [72]:

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```
data = {"name":"Reeta","age":20}
del data["name"]
print(data)
```

```
{'age': 20}
```

clear()

In [75]:

```
data = {"name":"Reeta","age":20}
data.clear()
print(data)
```

```
{}
```

13.copy

In [80]:

```
data = {"name":"Reeta","age":20}
res=data.copy()
print(data)
print(res)
```

```
{'name': 'Reeta', 'age': 20}
{'name': 'Reeta', 'age': 20}
```

In [82]:

```
data = {"name":"Reeta","age":20}
res=dict(data)
print(res)
```

```
{'name': 'Reeta', 'age': 20}
```

14.nested dict

In [87]:

```
sub={"Tamil":{"std1":99,"std2":56},
     "English":{"std1":96,"std2":68},
    "maths":{"std1":45,"std2":66}
}
print(sub)
```

```
{'Tamil': {'std1': 99, 'std2': 56}, 'English': {'std1': 96, 'std2': 68}, 'maths': {'std
1': 45, 'std2': 66}}
```

In [91]:

```
sub={"Tamil":{"std1":99,"std2":56},
     "English":{"std1":96,"std2":68},
    "maths":{"std1":45,"std2":66}
}
print(sub["Tamil"]["std1"])
```

```
99
```

15.fromkeys()

In [98]:

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```
name=("a","b")
age=20
a=dict.fromkeys(name,age)
print(a)
```

```
{'a': 20, 'b': 20}
```

In [100]:

```
name=("a","b")
a=dict.fromkeys(name)
print(a)
```

```
{'a': None, 'b': None}
```

In [102]:

```
name=("a","b")
age=20,22
a=dict.fromkeys(name,age)
print(a)
```

```
{'a': (20, 22), 'b': (20, 22)}
```

16.setdefault()

In [105]:

```
car = {
  "brand": "Ford",
  "model": "Mustang",
  "year": 1964
}

x = car.setdefault("model", "Bronco")

print(x)
```

```
Mustang
```

In [109]:

```
car = {
  "brand": "Ford",
  "model": "Mustang",
  "year": 1964
}

x = car.setdefault("color","red" )

print(x)
```

```
red
```

Practice Question

1. Write a Python program to create a dictionary containing student details and print it.

In [113]:

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```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
print(data)
```

```
{'name': 'Reeta', 'age': 20, 'city': 'Salem', 'grade': 'B'}
```

2. Write a Python program to get user input for a student's name and age and create a dictionary.

In [118]:

```
student={}
name=input("enter name:")
age=int(input("enter age:"))
student["name"]=name
student["age"]=age
print(student)
```

```
{'name': 'Reeta', 'age': 20}
```

3. Write a Python program to access a value using its key.

In [123]:

```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
res=data["name"]
print(res)
```

```
Reeta
```

4. Write a Python program to add a new key-value pair to an existing dictionary.

In [126]:

```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
data["class"]="A class"
print(data)
```

```
{'name': 'Reeta', 'age': 20, 'city': 'Salem', 'grade': 'B', 'class': 'A class'}
```

5. Write a Python program to update a student's age in a dictionary.

In [129]:

```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
data["age"]=21
print(data)
```

```
{'name': 'Reeta', 'age': 21, 'city': 'Salem', 'grade': 'B'}
```

6. Write a Python program to remove a key using the pop() method.

In [132]:

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```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
data.pop("grade")
print(data)
```

```
{'name': 'Reeta', 'age': 20, 'city': 'Salem'}
```

7. Write a Python program to access a value using the get() method when the key may not exist.

In [135]:

```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
a=data.get("name")
print(a)
```

```
Reeta
```

8. Write a Python program to use setdefault() to add a missing key to a dictionary.

In [142]:

```
data={"name":"Reeta","age":20,"city":"Salem","grade":"B"}
a=data.setdefault("class","A")
print(a)
print(data)
```

```
A
{'name': 'Reeta', 'age': 20, 'city': 'Salem', 'grade': 'B', 'class': 'A'}
```

9. Write a Python program to merge two dictionaries using the update() method.

In [147]:

```
a={"name":"Reeta","age":20}
b={"city":"Salem","grade":"B"}
a.update(b)
print(a)
```

```
{'name': 'Reeta', 'age': 20, 'city': 'Salem', 'grade': 'B'}
```

10. Write a Python program to create a dictionary using the fromkeys() method without using loops.

In [150]:

```
a=("name","age")
b="reeta",20
c=dict.fromkeys(a,b)
print(c)
```

```
{'name': ('reeta', 20), 'age': ('reeta', 20)}
```

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